



SECP2523
DATABASE (WBL)
SECTION 02

ASSIGNMENT 1

LECTURER:
DR. SEAH CHOON SEN

GROUP MEMBER:
TAN ZHI MING (A23CS0189)

QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	Order	Customer
Information	lphone	DHESH

QUESTION 2

Answer the questions below based on the following relational schema:

```
SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)
EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)
PROJECT (Proj_Code, Proj_Name, Proj_StartDate)
SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
```

- a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	-
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	-
SCIENTIST_EXPERIMENT	-	Exp_No, Sc_ID

- b) Does the table exhibit entity integrity? Answer ‘Yes’ or ‘No’ and then explain your answer. (4M)

TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	There has a primary key
EXPERIMENT	Yes	There has a primary key
PROJECT	Yes	There has a primary key

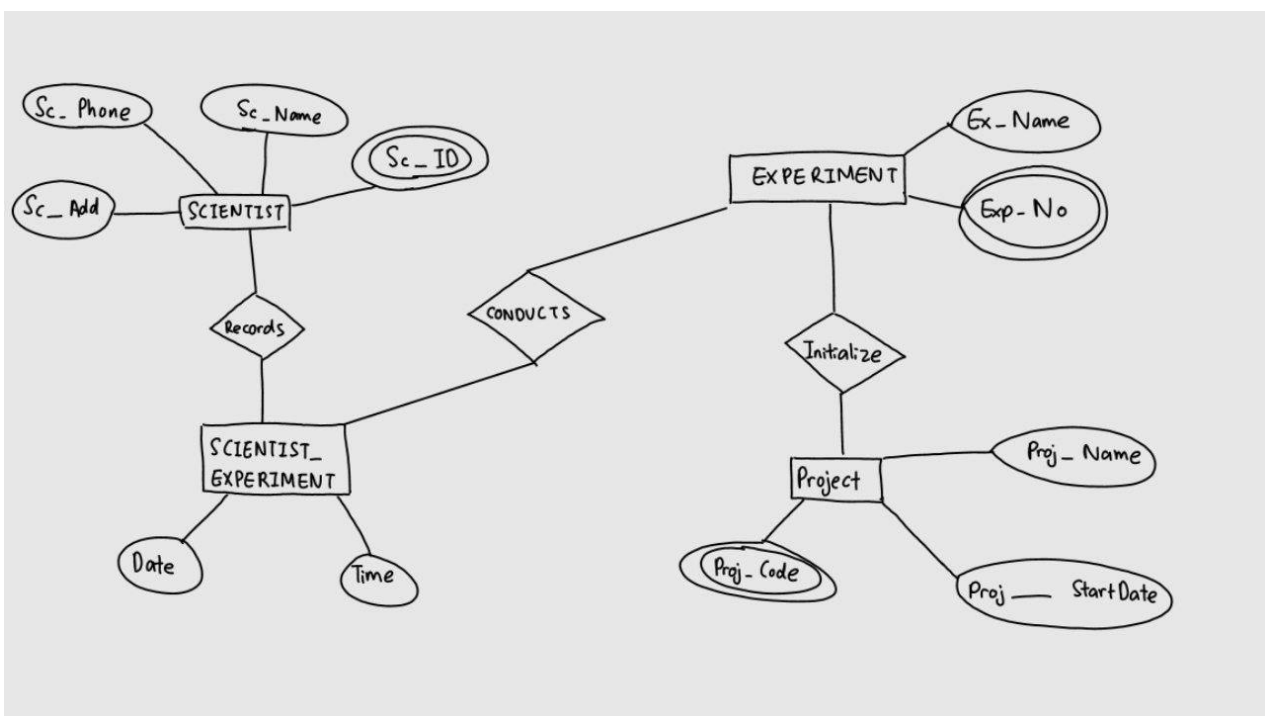
ASSIGNMENT 1 SECP2523 DATABASES

SCIENTIST_EXPERIMENT	No	It is NULL at primary key
----------------------	----	---------------------------

- c) Does the table exhibit referential integrity? Answer 'Yes' or 'No' and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY	EXPLANATION
SCIENTIST	No	The Foreign Key is NULL
EXPERIMENT	Yes	The Foreign Key match a primary key in the referenced table
PROJECT	No	The Foreign Key is NULL
SCIENTIST_EXPERIMENT	Yes	The Foreign Key match a primary key in the referenced table

- d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

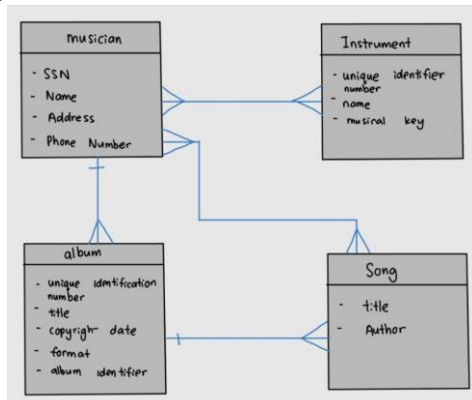
Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

Task:

1. List each Entity that is identified in Luxury Records System. **Musician** , **Instrument** , **Album** , **Song**
2. For each entity list the attributes.
Musician(SSN , name , address , phone)
Instrument(instrument_ID, instrument_name, instrument_key)
Album(album_ID, album_title, copyright_Date, album_format, album_Identifier)
Song(song_title , song_author)
3. Draw a logical model to represent each entity and the relationships between them.
4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.

3/4



QUESTION 4

Give an example of each of the three types of relationships.

- i) One-to-many relationship (one man can own many car)
- ii) Many-to-many relationship (Many customer can buy many product)
- iii) One-to-one relationship (Each student can only register a program)
- iv) Many-to-one relationship (many car can own by a man)

QUESTION 5

CLIENT (Client_Number, Client_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)

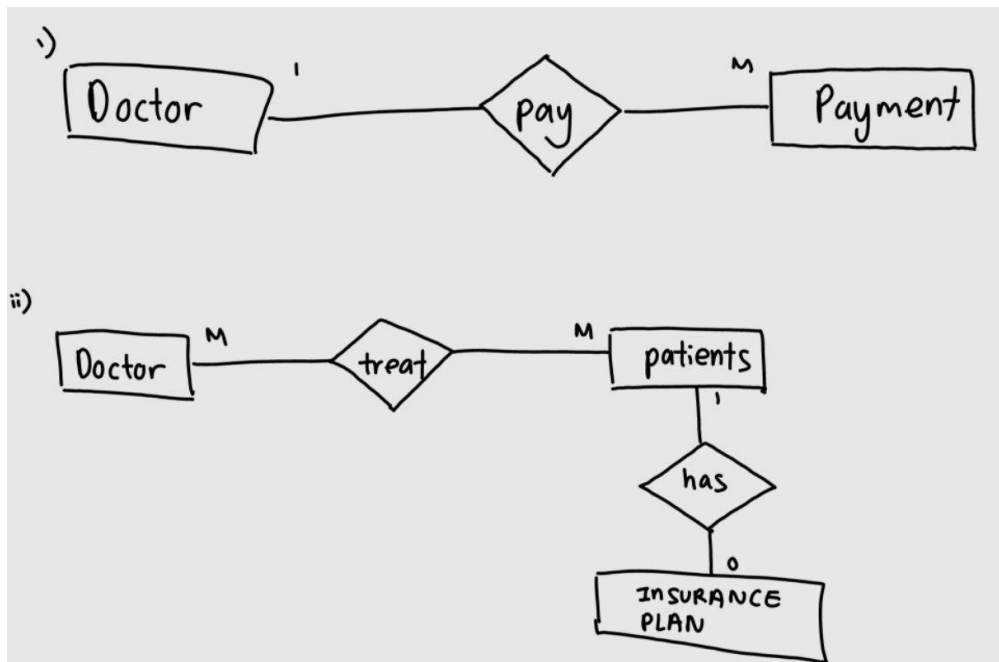
RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

- a) What is the relationship between Recruiter and Client?(front first) (Many-to-one relationship)
- b) What is the primary key for the Client table and Recruiter table?
Client(Client_Number) , Recruiter(Recruiter_Number)
- c) Define foreign key? Identify foreign key in the Client table.
Client(Recruiter_Number)

QUESTION 6

Draw the Chen & Crow's Foot notation based on the following statements:

- i) "A customer can make many payments, but each payment is made by only one customer."
- ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(SSN, Name, Major, Bdate)

COURSE(Course#, Cname, Dept)

ENROLL(SSN, Course#, Quarter, Grade)

BOOK_ADOPTION(Course#, Quarter, Book_ISBN)

TEXT(Book ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema.

